

Karachi Food Delivery Time Analysis — Auto Analysis

Business Analytics Report

Project: Karachi Food Delivery Time Analysis

Dataset: karachi_food_delivery_dataset.csv

Analysis date: 2026-09-25

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **karachi_food_delivery_dataset.csv** (2,500 rows, 18 columns) covering 2025-07-01 to 2026-06-30 to answer the business question: *A synthetic dataset of 2,520 food delivery orders across eight Karachi areas (Jul 2025 to Jun 2026), with 18 columns covering order time, area, cuisine, weather, traffic, vehicle type, distance, items, order value, preparation time, rider rating and experience, delivery time, promised time, late delivery, tips, and customer rating. It includes missing values, duplicates, mixed data types, inconsistent text, impossible values, and outliers for testing data cleaning. Suitable for delivery time regression and late delivery classification..*

Key verified findings

- The original dataset contained 533 missing value(s) and 20 duplicate row(s), which have since been cleaned.
- A Gradient Boosting model was trained to predict **Late Delivery** (classification). The model correctly classifies about 91.0% of cases (Accuracy).
- 2 features excluded as data leakage before training (Delivery Time Min, Promised Time Min) to avoid an artificially inflated score.

Model reliability, in plain terms: The model correctly classifies about 91.0% of cases (Accuracy). F1 score (balancing precision and recall) is 0.907. ROC-AUC is 0.959 (1.0 = perfect separation, 0.5 = no better than chance).

Top recommendations

- The most common observed outcome for Late Delivery is 'No' (375 of 500 test cases).
- Among records, the 'Distance Km: Higher' group shows the highest observed rate (37.2%, n=834), compared to 'Distance Km: Lower' (8.6%, n=833).

2. About the Dataset

karachi_food_delivery_dataset.csv contains 2,500 rows and 18 columns. It spans the period 2025-07-01 to 2026-06-30 (based on Order Time).

Business question: A synthetic dataset of 2,520 food delivery orders across eight Karachi areas (Jul 2025 to Jun 2026), with 18 columns covering order time, area, cuisine, weather, traffic, vehicle type, distance, items, order value, preparation time, rider rating and experience, delivery time, promised time, late delivery, tips, and customer rating. It includes missing values, duplicates, mixed data types, inconsistent text, impossible values, and outliers for testing data cleaning. Suitable for delivery time regression and late delivery classification.

Column dictionary

Column	Type	Missing	Notes
Order Id	Identifier	0.0%	
Order Time	Identifier	0.0%	
Area	Category	0.0%	most common: DHA
Cuisine	Category	0.0%	most common: Desserts
Weather	Category	0.0%	most common: Clear
Traffic Level	Category	0.0%	most common: Medium
Vehicle Type	Category	0.0%	most common: Motorbike
Distance Km	Numeric	0.0%	typical range 0.73–22.01
Items Count	Numeric	0.0%	typical range 1.00–8.00
Order Value Pkr	Numeric	0.0%	typical range 351.00–7,115.
Prep Time Min	Numeric	0.0%	typical range 5.00–49.00
Rider Rating	Numeric	0.0%	typical range 3.30–5.00

Rider Experience Months	Numeric	0.0%	typical range 0.00–59.00
Delivery Time Min	Numeric	0.0%	typical range 10.00–354.00
Promised Time Min	Numeric	0.0%	typical range 55.00–70.00
Late Delivery	Category	0.0%	most common: No
Tip Pkr	Numeric	0.0%	typical range 0.00–200.00
Customer Rating	Numeric	0.0%	typical range 1.00–5.00

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
2,520	533	20	13

Duplicate Rows

Column	Finding	Affected
(whole dataset)	20 duplicate row(s) detected.	20

Missing Values

Column	Finding	Affected
order_time	'order_time' has 5 missing value(s) (0.2%).	5
weather	'weather' has 101 missing value(s) (4.0%).	101
traffic_level	'traffic_level' has 75 missing value(s) (3.0%).	75
prep_time_min	'prep_time_min' has 75 missing value(s) (3.0%).	75
rider_rating	'rider_rating' has 126 missing value(s) (5.0%).	126
customer_rating	'customer_rating' has 151 missing value(s) (6.0%).	151

Outliers

Column	Finding	Affected
prep_time_min	'prep_time_min' has 1 statistical outlier(s) outside [3.5, 47.5].	1
rider_rating	'rider_rating' has 1 statistical outlier(s) outside [3.35, 5.35].	1
delivery_time_min	'delivery_time_min' has 81 statistical outlier(s) outside [4.5, 96.5].	81
tip_pkr	'tip_pkr' has 391 statistical outlier(s) outside [-75.0, 125.0].	391
customer_rating	'customer_rating' has 38 statistical outlier(s) outside [2.5, 6.5].	38

Impossible Values

Column	Finding	Affected
delivery_time_min	'delivery_time_min' has 3 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	3

Cleaning performed

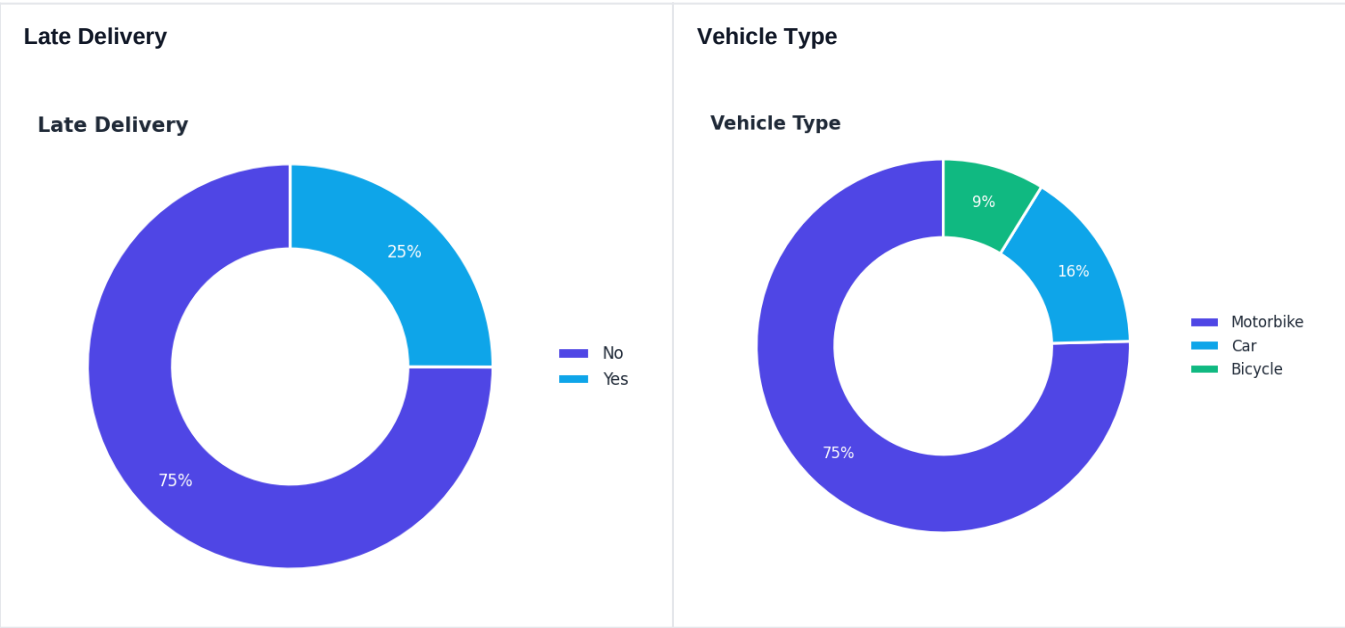
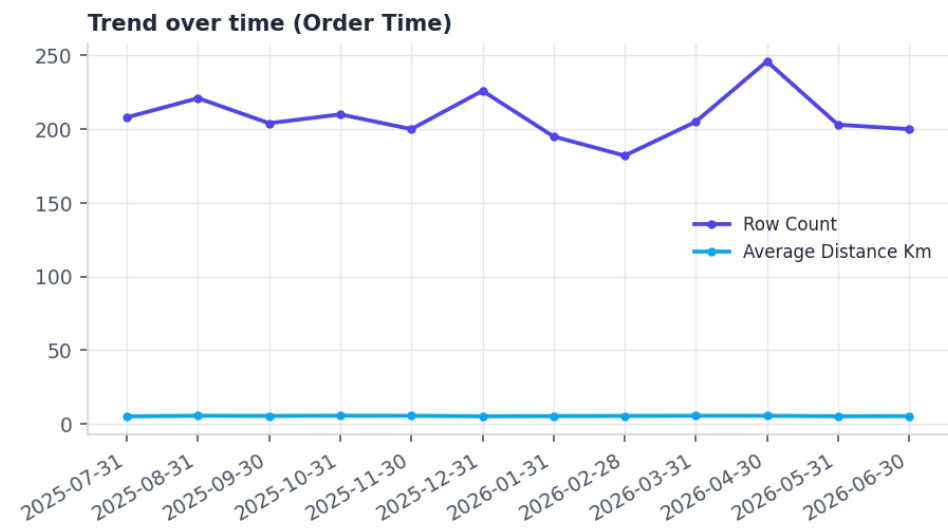
Problem	Before	Action	After	Reason
Duplicate rows	20 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Order Time	5 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Weather	101 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Traffic Level	75 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Missing Prep Time Min	75 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Rider Rating	126 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Missing Customer Rating	151 missing	Filled using the column median	0 missing	Reduces the influence of extreme values compared to the mean.
Impossible negative Delivery Time Min	3 impossible value(s)	Converted to missing, then filled	0 impossible values	A rare negative value in an otherwise non-negative column is almost always a data-entry error.
Text-formatted numbers in Distance Km	stored as text with units (e.g. '4.2 km')	Converted to a clean numeric value	numeric	100% of values matched a number-with-unit pattern.
Inconsistent capitalization in Weather	5 variant label(s) (e.g. 'rainy' vs 'Rainy')	Merged into a single canonical label	consistent labels	Values differing only by letter case would otherwise be treated as different categories.

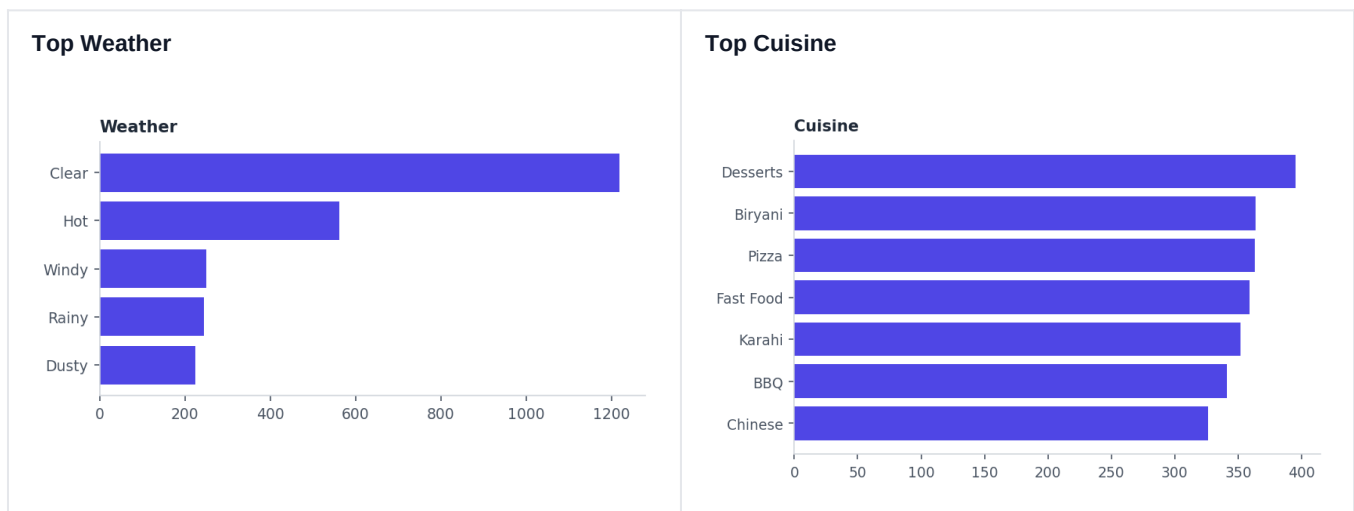
After cleaning: 0 missing cell(s), 0 duplicate row(s), 2,500 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Late Delivery Rate	Total Distance Km	Total Items Count
2,500	25.0%	13,095	10,969

Trend over time (Order Time)





Items Count breakdown

Average: 4.39 (range 1.0–8.0)

Value	Count	Percent
8	292	11.7%
7	306	12.2%
6	302	12.1%
5	293	11.7%
4	292	11.7%
3	336	13.4%
2	359	14.4%
1	320	12.8%

5. Modeling Approach

A **classification** model was trained to predict **Late Delivery**, using 14 feature(s).

Features used

Order Time: Year, Order Time: Month, Order Time: Dayofweek, Area, Cuisine, Weather, Traffic Level, Vehicle Type, Distance Km, Items Count, Order Value Pkr, Prep Time Min, Rider Rating, Rider Experience Months

Features excluded, and why

Column	Reason excluded
Order Id	Identifier column (e.g. an ID) — carries no predictive signal.
Order Time	Raw date — replaced with derived year / month / day-of-week features instead.
Delivery Time Min	Leakage. Together with 'promised_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns.
Promised Time Min	Leakage. Together with 'delivery_time_min', this column almost exactly reproduces the target (99.8% agreement via a simple comparison) — the model could 'cheat' by combining the two instead of learning real patterns.
Tip Pkr	Excluded by default 'tip_pkr' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (34% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.
Customer Rating	Excluded by default 'customer_rating' often represents information only available AFTER the outcome occurs (e.g. a rating or tip given once an order is complete), and shows a real (46% correlated) relationship with the target here. Review whether this would actually be known at prediction time — if not, exclude it below.

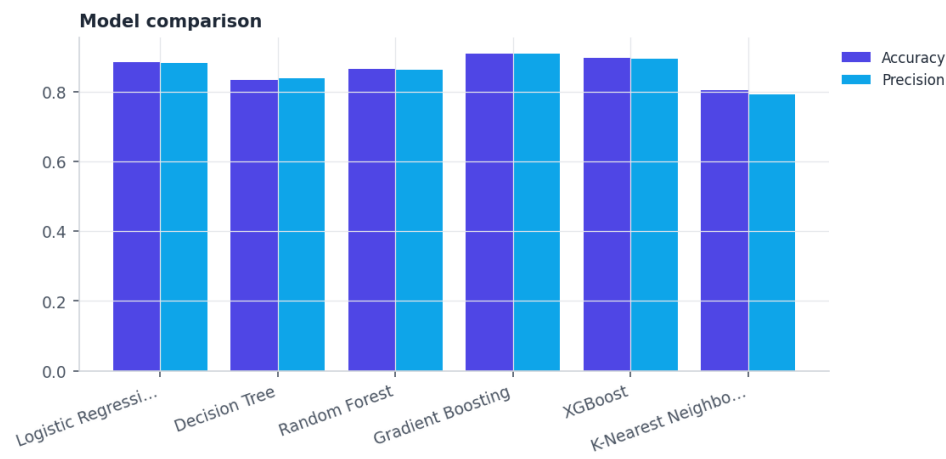
Technical details

Algorithm: Gradient Boosting · Preprocessing: 9 numeric + 5 categorical feature(s), scaled/encoded · Train/test split: 20.0% held out · Cross-validation folds: 5 · Random seed: 42

6. Model Performance

Model comparison

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.8860	0.8829	0.8860	0.8831	0.9377
Decision Tree	0.8340	0.8384	0.8340	0.8359	0.7907
Random Forest	0.8660	0.8640	0.8660	0.8563	0.9403
Gradient Boosting	0.9100	0.9090	0.9100	0.9068	0.9591
XGBoost	0.8960	0.8936	0.8960	0.8939	0.9458
K-Nearest Neighbors	0.8040	0.7927	0.8040	0.7768	0.8120

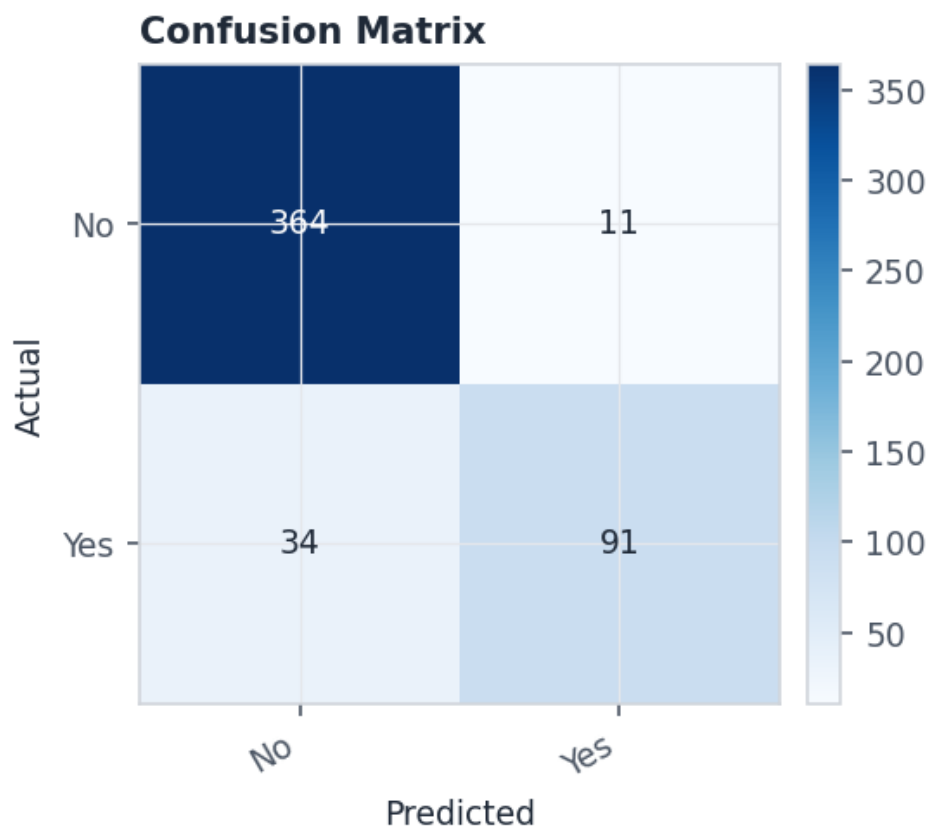


In plain language

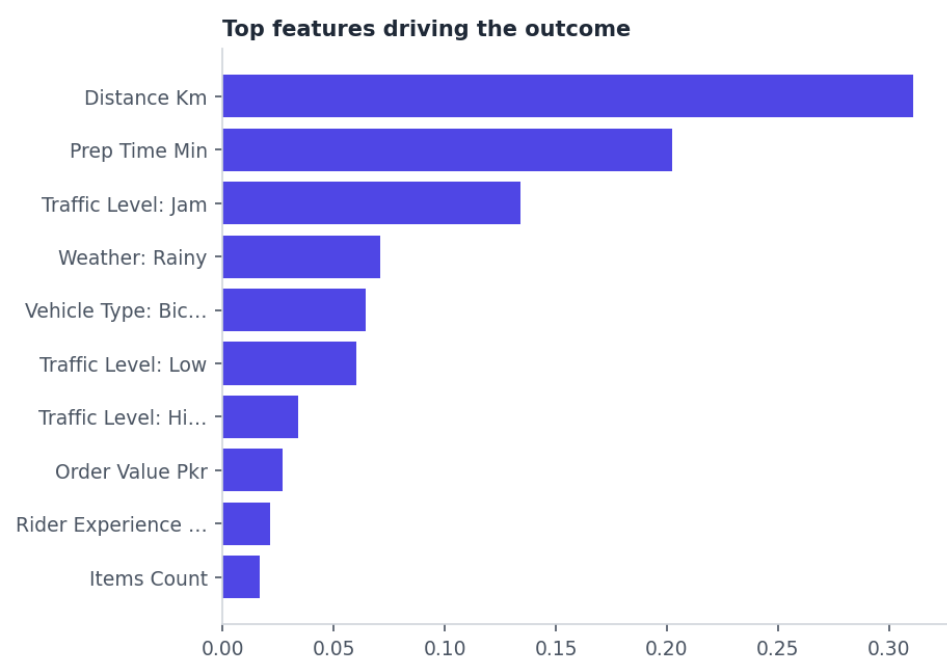
- The model correctly classifies about 91.0% of cases (Accuracy).
- F1 score (balancing precision and recall) is 0.907.
- ROC-AUC is 0.959 (1.0 = perfect separation, 0.5 = no better than chance).

Compared against a trivial baseline (F1 Score = 0.6429): this model clearly outperforms it.

Confusion Matrix



7. What Drives the Outcome



Feature	Relative importance
Distance Km	31.1%
Prep Time Min	20.2%
Traffic Level: Jam	13.4%
Weather: Rainy	7.1%
Vehicle Type: Bicycle	6.4%
Traffic Level: Low	6.0%
Traffic Level: High	3.4%
Order Value Pkr	2.7%
Rider Experience Months	2.1%
Items Count	1.7%

8. Future Outlook

- The most common observed outcome for Late Delivery is 'No' (375 of 500 test cases).
- Among records, the 'Distance Km: Higher' group shows the highest observed rate (37.2%, n=834), compared to 'Distance Km: Lower' (8.6%, n=833).

Group	Observed rate	Group size
Distance Km: Lower	8.6%	833
Distance Km: Middle	29.3%	833
Distance Km: Higher	37.2%	834

9. AI Business Insights AI-generated

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- The following column(s) were excluded as likely data leakage and could not be used: `delivery_time_min`, `promised_time_min`. If any of these should genuinely be available before prediction time, review them manually.
- Cross-validation scores varied by ± 0.016 across folds — treat the headline metric as an estimate with this much natural uncertainty, not an exact figure.
- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Full model metrics

Metric	Value
Accuracy	0.91
Precision	0.909
Recall	0.91
F1 Score	0.9068
ROC-AUC	0.9591
Cv Mean	0.8917
Is Weak	False

Preprocessing

Setting	Value
Test Size	0.2
Cv Folds	5
Random seed	42
Model version	4

Hyperparameters

Parameter	Value
Random Seed	42

Dataset statistics

Statistic	Value
Rows	2,500
Columns	18